

Jessica Gagliardi (she/her/hers)
PhD Candidate
University of California, Santa Cruz
jgagliardi@ucsc.edu

Education

University of California, Santa Cruz
Doctoral studies, Earth & Planetary Science
Advisor: Prof. Terrence Blackburn

University of California, Berkeley
B.A. Geophysics, 2020

Honors Thesis: Timing and patterns of glacial incision along the eastern Antarctic Peninsula: Insights from low-temperature thermochronometry (Advisors: Prof. David Shuster and Dr. Anna Clinger)

Research Interests

Geochronology, geochemistry, paleohydrology of the Antarctic ice sheet, U-Th dating, paleoclimate, novel paleo-proxies, subglacial chemical precipitates

Honors, Awards & Certificates

ARCS Scholar, Achievement Rewards for College Scientists Foundation (2025)
Graduate Pedagogy Fellow, UCSC Teaching & Learning Center (2025)
J. Casey Moore Award, UCSC EPS Department (2024)
Equity-Minded Mentoring Certification, UCSC Teaching & Learning Center (2024)
West Antarctic Ice Sheet Workshop Early Career Travel Grant (2023)
Dean's Research Travel Grant, UC Santa Cruz (2021, 2022, 2023)
Regent's Fellowship, UC Santa Cruz (2021)

First-Author Publications

J. Gagliardi, G. Piccione, T. Hillebrand, T. Blackburn, S. Tulaczyk. Subglacial precipitates from a Ross Sea outlet glacier show evidence of early Antarctic ice sheet response to ocean warming after Last Glacial Maximum (in prep).

J. Gagliardi, T. Blackburn, G. Piccione, S. Tulaczyk, C.B. Keller. Subglacial precipitates record Antarctic ice sheet response to Southern Ocean warming. *Geophysical Research Letters* (2025).

Coauthored Publications

G. Piccione, T. Blackburn, M.P. Hain, S. Tulaczyk, C. Carney, **J. Gagliardi**. Microbes drive subglacial CO₂ production and silicate weathering throughout Antarctica (in review).

S. Kodama, T. Pico, A. Robel, J. Christian, N. Gomez, C. Vigilia, E. Powell, **J. Gagliardi**, S. Tulaczyk, T. Blackburn. Glacial isostatic adjustment modulates potential grounding line stability zones in the Ross Sea (Antarctica) since the Last Glacial Maximum (2024).

Conference Abstracts & Talks

J. Gagliardi, T. Blackburn, S. Tulaczyk, G. Piccione. Subglacial precipitates from a Ross Sea outlet glacier show evidence of early Antarctic ice sheet response to ocean warming after Last Glacial Maximum. Oral Presentation. Goldschmidt Conference, Prague, Czechia. Abstract: 32296. 2025

J. Gagliardi, T. Blackburn, S. Tulaczyk, G. Piccione. Subglacial precipitates constrain sequence of post-LGM ice lowering in a Ross Sea outlet glacier. Oral Presentation. American Geophysical Union Fall Meeting. Abstract: PP11B-01. 2024

J. Gagliardi, T. Blackburn, S. Tulaczyk, G. Piccione. Subglacial precipitates constrain timing and forcing of ice lowering in a Ross Sea outlet glacier during Termination I. Oral Presentation. West Antarctic Ice Sheet Workshop. 2023

J. Gagliardi, T. Blackburn, S. Tulaczyk, G. Piccione, C. B. Keller. Subglacial Chemical Precipitates Record Antarctic Ice Sheet Response to Southern Ocean Warming. Oral presentation. Goldschmidt Conference, Lyon, France. Abstract: 15024. 2023

J. Gagliardi, S. Kodama, T. Pico, C. Vigilia, A. Robel, E. M. Powell, S. M. Tulaczyk, T. Blackburn. Identifying pinning point elevation thresholds for Ross Sea ice streams (West Antarctica) over the last deglaciation based on glacial isostatic adjustment and grounding line simulations. American Geophysical Union Fall Meeting. Abstract: PP52D-0458. 2022

J. Gagliardi, T. Blackburn, S. Tulaczyk, G. Piccione. Antarctic Subglacial Precipitates Record Ice Sheet Response to Suborbital Southern Ocean Warming. Oral presentation. Graduate Climate Conference, Pack Forest, WA. Abstract: CLI_099. 2022

J. Gagliardi, T. Blackburn, S. Tulaczyk, G. Piccione. Antarctic subglacial precipitates record ice sheet response to millennial-scale Southern Ocean warming. Oral presentation. West Antarctic Ice Sheet Workshop, Estes Park, CO. 2022

J. Gagliardi, T. Blackburn, S. Tulaczyk, G. Piccione. Antarctic subglacial precipitates record ice sheet response to millennial-scale Southern Ocean warming. Oral presentation. Scientific Committee on Antarctic Research Open Science Conference, Virtual. Abstract: 641. 2022

J. Gagliardi, T. Blackburn, S. Tulaczyk, G. Piccione. Antarctic subglacial precipitates record ice surging in response to millennial-scale Southern Ocean warming. American Geophysical Union Fall Meeting, New Orleans, LA. Abstract: C25E-0837. 2021

J. Gagliardi, A. Clinger, D. Shuster. Timing and patterns of glacial incision along the eastern Antarctic Peninsula: Insights from low-temperature thermochronometry. Oral presentation. American Geophysical Union Fall Meeting. Abstract: EP026-07. 2020

Research Experience

Graduate Student Researcher, Blackburn Lab, UC Santa Cruz Department of Earth & Planetary Science (July 2021-present)

Research Assistant and Lab Technician for Antarctic Erosion Project, Shuster Lab, UC Berkeley Department of Earth & Planetary Sciences (May 2019–December 2020)

Research Apprentice, UC Berkeley Undergraduate Research Apprenticeship Program (September 2017–May 2019)

Transcriber & Research Assistant, Mark Twain Papers & Project (June 2018–May 2019)

Teaching & Mentoring Experience

Glaciology (EART 148), TA, Spring 2023, Winter 2026

Introduction to Pedagogy (EART 203), Instructor of Record, Fall 2025

Equitable Course Design (GRAD 215), UCSC Teaching & Learning Center, Student, Spring 2025

Earth as a Chemical System (EART 110B), TA, Winter 2024

History & Evolution of the Earth (EART 110A), TA, Fall 2022, Fall 2023 & Fall 2024

Undergraduate Mentor (5 students), Blackburn Lab, 2021-present

Community Service & Outreach

Science Communication & Public Outreach Volunteer, Santa Cruz Museum of Natural History (2021-present)

UCSC GEODES (Geologists Encouraging Openness & Diversity in Earth Science) Organizing Committee (2021-present)

Graduate Student Representative to Faculty, UCSC EPS (2023-2024)